Does a Learned Tokenizer Help a Small DNA Language Model? GBST-BERT on the Human Genome

Omar Mohamed (K12103822)
Practical Work, Institute for Machine Learning
Johannes Kepler University Linz, Austria

Abstract

DNA language models usually fix their tokenization in advance. ChaRNABERT showed for RNA that a learned tokenizer, Gradient-Based Subword Tokenization (GBST), can replace this choice. We implement this architecture in the DNA-xLSTM code base and pretrain a 2.1M-parameter model, GBST-BERT, on the human genome under the short-context protocol of Bio-xLSTM, next to a Transformer++ control of the same shape. Pretraining does not separate the two models (test accuracy on masked bases 56.44% vs. 56.33%). On the 18 Nucleotide Transformer tasks, GBST-BERT has the best mean of seven models on 13 tasks and exceeds the control on 16, with the largest gains on histone marks. On the 8 Genomic Benchmarks tasks the advantage is small (higher mean on 5 of 8). GBST places 98% of its weight on single bases, so it does not learn multi-base units on human DNA; we suspect the gain comes from its convolution, but the control also differs in normalization and learning-rate tuning, so this remains to be tested. Finally, we show that the weak enhancer scores coincide with a train–test mismatch in the benchmark itself: a plain 4-mer classifier shows the same drop from validation to test, and a quarter of the test sequences are exact copies of training sequences.

Ten questions and answers

1. What question does this work ask? Whether a learned tokenizer helps a small DNA language model, compared with reading one base at a time, at the 2M-parameter scale of DNA-xLSTM.

2. What is GBST, and why try it on DNA? GBST builds candidate blocks of 1 to 4 bases around each position and learns how much weight each one receives. Functional DNA units have varied lengths, from CpG pairs to codons and motifs, so a model that chooses its own units is attractive (Section 1).

3. What was built? GBST-BERT: GBST followed by a bidirectional 4-block Transformer encoder, 2,107,392 parameters, added to the DNA-xLSTM code base. It comes with 26 automated tests, a one-file version of the model and a notebook covering the whole pipeline (Section 3).

4. How was it trained? By masked language modelling on hg38, with the short-context protocol of Bio-xLSTM: 10,000 steps of 2^{20} bases, 10.5B bases in total (Section 4).

5. How was it evaluated? On masked-base prediction, on the 18 Nucleotide Transformer tasks (10 seeds) and on the 8 Genomic Benchmarks tasks (5 seeds), next to a Transformer++ control and the values published in Bio-xLSTM. No choice was made on test data.

6. What is the main result? Pretraining is a tie (56.44% vs. 56.33%). On the NT tasks GBST-BERT has the best mean of seven models on 13 of 18 and exceeds its control on 16 of 18. On Genomic Benchmarks the advantage is small: a higher mean on 5 of 8 tasks.

Preprint.

7. Did GBST learn DNA "words"? No. The final model places 98% of its weight on single bases, so it does not form multi-base units on human DNA in this setup (Section 5.2).

8. Does this prove that the tokenizer causes the gain? No. The control also differs in normalization and in how its learning rate was chosen, and each model was pretrained once. We suspect GBST's convolution, but that is untested; replacing GBST with an embedding table in the same model would settle it (Section 6).

9. Why are enhancers the weakest tasks? Both enhancer tasks score far lower on test than on validation, unlike the other 16 tasks, and a plain 4-mer classifier shows the same drop. This points to a train–test mismatch in the dataset. A quarter of the test sequences are also exact copies of training sequences (Section 5.4).

10. Can the results be reproduced, and what comes next? The notebook re-computes all tables and figures from the saved runs and repeats one fine-tuning job per benchmark; switches repeat the long runs. The next steps are the ablation in question 8 and repeated pretraining with other seeds.

1 Introduction

Genomes are long texts over a four-letter alphabet, and language models trained on them have become a general tool for predicting regulatory and functional properties of DNA [Dalla-Torre et al., 2023, Nguyen et al., 2023, Schiff et al., 2024]. Before such a model can learn anything, it must decide how to cut its input into units, and for DNA this choice is not obvious. Natural language offers words as a starting point; DNA has no spaces, and its functional units have very different lengths and meanings. A methylation site involves two bases, a codon three, and a transcription-factor binding site typically six to ten; splice sites depend on short dinucleotides in a wider context, and the same sequence can matter in one region and not in another.

Fixed tokenizations have known trade-offs. One token per base keeps single-nucleotide resolution, which matters for variants, and is used by HyenaDNA, Caduceus and DNA-xLSTM, but it leaves all grouping of bases to the network. K-mers, as in the Nucleotide Transformer, shorten the sequence, but they fix the unit length in advance, and a single base change or shift alters every token that overlaps it, so that the same motif can look different depending on where it falls relative to token boundaries. In both cases the unit is chosen by the modeller before seeing the data.

A learned alternative. GBST [Tay et al., 2022] makes this choice part of training. At each position it forms candidate blocks of several sizes and learns how much weight each deserves, so that gradients shape the tokenization together with the rest of the network. ChaRNABERT [Morales-Pastor et al., 2024] removed GBST's downsampling step to keep one output per character, combined it with a bidirectional encoder, and trained RNA models of 8M to 650M parameters on 62 million sequences. It reports a lower loss than single-character input in most settings, for example 0.498 vs. 0.508 for its 5M model. If such a module also helps on DNA, it would be a cheap addition: in our model it accounts for 3.3% of the parameters.

Why the small-model setting. DNA-xLSTM [Schmidinger et al., 2024] shows that 2M-parameter models can be strong on DNA, and it publishes a complete protocol: data, training budget, fine-tuning procedure, and scores of several architectures. This gives a fixed reference. A small model is also where a good inductive bias should matter most, and where repeated fine-tuning remains affordable on shared GPUs.

Research questions. We keep ChaRNABERT's architecture and change the data, and we ask:

- **RQ1.** Does GBST improve masked-base prediction on the human genome?
- **RQ2.** Does it improve downstream prediction, relative to a same-shape control and to published models of similar size?
- **RQ3.** What does GBST learn on DNA: does it form multi-base units?

Contributions.

- An implementation of GBST-BERT in the DNA-xLSTM code base, tested against a reference implementation of the published equations, and released with a one-file model and a notebook.

- A comparison with Transformer++ in pretraining and on 26 downstream tasks (464 fine-tuning runs), reported with all repeats.
- An analysis of what GBST learned on DNA, which differs from the motivation for using it.
- A diagnosis of the two NT enhancer tasks, which affects how these tasks should be read.

Section 2 gives background, Section 3 the method, Section 4 the data and protocols, Section 5 the results, and Section 6 what they do and do not support.

2 Background

Tokenization of biological sequences. Bio-xLSTM and Caduceus [Schiff et al., 2024] use one token per base, as does HyenaDNA [Nguyen et al., 2023]; the Nucleotide Transformer [Dalla-Torre et al., 2023] uses k-mers. GBST [Tay et al., 2022] forms candidate blocks of characters and lets a scoring network weight them, which makes the tokenization differentiable. ChaRNABERT removes Charformer's downsampling, so that one output per character remains.

Evaluation. Bio-xLSTM evaluates DNA models of 2M parameters on the 18 Nucleotide Transformer (NT) tasks, and of 500k parameters on the 8 Genomic Benchmarks tasks [Grešová et al., 2023]. We reuse its reported values as references.

3 Method

The model has three parts: GBST, a bidirectional encoder and a task head. Only GBST distinguishes it from a standard character-level encoder.

3.1 Input

A sequence is a list of L tokens $x_1, \ldots, x_L$ with one token per base (A = 7, C = 8, G = 9, T = 10, N = 11; 0–6 are special tokens such as `[MASK]` and `[PAD]`), using the repository's character tokenizer. The vocabulary has 12 entries. All candidate blocks are built inside the model, so no tokenizer needs to be trained separately.

3.2 GBST

GBST maps the tokens to one vector per position, $z_1, \ldots, z_L \in \mathbb{R}^d$ with $d = 256$, in five steps.

(1) Local mixing. Each token is embedded, passed through a depthwise convolution of width $M = 4$ (one filter per dimension, padded to keep the length), and then through a pointwise linear layer:

$$u = \mathrm{Pointwise}\big(\mathrm{DWConv}_M(E[x_1], \ldots, E[x_L])\big). \tag{1}$$

After this step, each position already holds information about its neighbours.

(2) Candidate blocks. For every block size $b \in \{1, \ldots, M\}$ and offset $o \in \{0, \ldots, b-1\}$, position i receives the block that starts o positions before it, pooled by a sum:

$$B_{i,(b,o)} = \sum_{j=i-o}^{i-o+b-1} u_j, \tag{2}$$

where positions outside the sequence contribute zero. This gives $K = \sum_{b=1}^{M} b = 10$ candidates per position, all containing position i (Table 1). Offsets act like reading frames.

(3) Scoring. A linear scorer $w \in \mathbb{R}^d$ rates every candidate, and a softmax over the K candidates gives weights $P \in \mathbb{R}^{L \times K}$:

$$P_{i,k} = \frac{\exp(w^\top B_{i,k})}{\sum_{k'} \exp(w^\top B_{i,k'})}. \tag{3}$$

(4) Calibration. Following ChaRNABERT (its Eq. 6), positions with similar weight patterns are pulled towards each other:

$$\hat{P} = \mathrm{softmax}\big(PP^\top\big)\, P, \tag{4}$$

Table 1: The 10 GBST candidates for position 3 (the first T, counting from 0) of ACGTTGCA.

Block size	Candidate blocks containing position 3
1	T
2	TT, GT
3	TTG, GTT, CGT
4	TTGC, GTTG, CGTT, ACGT

with the softmax taken over rows. This step costs $O(L^2K)$, which is modest at $L = 1{,}024$.

(5) Mixing. Position i receives the weighted sum of its candidates, $z_i = \sum_k \hat{P}_{i,k}\, B_{i,k}$. As in ChaRNABERT there is no downsampling, so the output has one vector per base and the encoder sees the full sequence length.

GBST has 70,144 parameters: embedding 3,072, convolution 1,280, pointwise layer 65,536, scorer 256. The pointwise layer is not in ChaRNABERT's equations; we infer it from the paper's parameter table (Appendix C). We set $M = 4$ ourselves, as the paper does not state the maximum block size.

3.3 Encoder and pretraining objective

The encoder is a stack of 4 pre-norm blocks with bidirectional attention:

$$h' = h + \mathrm{Attn}(\mathrm{LN}(h)), \qquad h_{\text{next}} = h' + \mathrm{SwiGLU}(\mathrm{LN}(h')). \tag{5}$$

Attention uses 8 heads, rotary position embeddings [Su et al., 2024] and LayerNorm on queries and keys (QK normalization). The feed-forward layer is SwiGLU [Shazeer, 2020] with hidden size 320. All norms and linear layers have no bias. A final LayerNorm and a linear head give a score for each vocabulary entry at each position. The whole model has 2,107,392 parameters.

Pretraining uses masked language modelling [Devlin et al., 2019]. A set $\mathcal{M}$ of 15% of the positions is selected; of these, 80% are replaced by [MASK], 10% by a random token, and 10% stay unchanged. With $\tilde{x}$ the corrupted sequence, the loss is the mean cross-entropy over the selected positions,

$$\mathcal{L} = -\frac{1}{|\mathcal{M}|} \sum_{i \in \mathcal{M}} \log p_\theta(x_i \mid \tilde{x}), \tag{6}$$

in nats, so that uniform guessing among four bases gives $\ln 4 = 1.386$. Masking is applied to the input tokens, before GBST, so a masked base also removes information from the blocks around it.

3.4 Control model

Transformer++-2M [Schmidinger et al., 2024, Table A3] serves as the control. Table 2 lists how the two models compare. It is a close control but not a perfect one, and the differences beyond GBST limit what can be concluded (Section 6).

Table 2: GBST-BERT and its control.

	GBST-BERT	Transformer++
Input layer	GBST (70,144 parameters)	embedding (3,072 parameters)
Encoder shape	4 blocks, width 256, 8 heads, hidden 320	same
Position, feed-forward	RoPE, SwiGLU	same
Normalization	LayerNorm + QK normalization	RMSNorm
Pretraining LR	2e−3, from our search	1e−2, from the paper
Total parameters	2,107,392	2,036,992

3.5 Fine-tuning

For each downstream task, the MLM head is replaced by a new linear classifier on the mean of the final hidden states, $\bar{h}(x) = \frac{1}{L}\sum_i h_i(x)$, and all weights are trained with cross-entropy. Because DNA is double-stranded, we use post-hoc conjoining (PH) [Schiff et al., 2024]: in training each

sequence is replaced by its reverse complement $\mathrm{rc}(x)$ with probability 0.5, and at validation and test time the logits of both strands are averaged,

$$\hat{y}(x) = \tfrac{1}{2}\big(W\bar{h}(x) + c\big) + \tfrac{1}{2}\big(W\bar{h}(\mathrm{rc}(x)) + c\big). \tag{7}$$

3.6 Implementation and tests

The model is in `models/charna/`, registered in the repository's training program; existing models are unchanged. A launcher runs the code on GPUs that cannot build `mamba_ssm` and `flash_attn`. All 26 automated tests pass. Among other things, they compare GBST with a slow reference implementation of the equations, check the parameter count against ChaRNABERT's Table 9, verify that information flows in both directions, that the loss counts only selected positions, and that the one-file model `gbst_bert.py` matches the training code exactly. The model also memorized a single batch (loss below 0.05 after 164 steps), which confirms that it can fit data.

4 Experimental setup

Pretraining data. We use hg38 as prepared by the DNA-xLSTM repository: chromosomes 1–22 and X, 3.03B bases, 4.0% of them unknown (N). A BED file assigns 38,171 regions to the splits; each is widened to 2^{20} bases and cut into pieces of 1,024 bases. This gives 34.8M training pieces (2.33B distinct bases), 2.3M validation pieces (0.29B) and 2.0M test pieces (0.27B); regions of different splits do not overlap.

Pretraining. We follow the short-context protocol of Bio-xLSTM: pieces of 1,024 bases, 2^{20} bases per step, 10,000 steps (10.5B bases, about 4.5 passes over the training bases). A sequence is replaced by its reverse complement with probability 0.5. For GBST-BERT, learning rates $\{5\mathrm{e}{-4}, 1\mathrm{e}{-3}, 2\mathrm{e}{-3}, 5\mathrm{e}{-3}\}$ were compared over 500 steps by validation loss, and $2\mathrm{e}{-3}$ was used; Transformer++ keeps the paper's $1\mathrm{e}{-2}$. Each model was pretrained once. Appendix A lists all settings.

NT benchmark. The 18 tasks [Dalla-Torre et al., 2023] cover 10 histone marks, 2 enhancer tasks, 3 promoter tasks and 3 splice-site tasks, with 200 to 600 bases per sequence (Appendix B). Following Schmidinger et al. [2024, App. C.5], we fine-tune as in Section 3.5 for 20 epochs and test the epoch with the best validation score, with 10 seeds, each drawing a new 90/10 train/validation split. Scores follow the paper: Matthews correlation (MCC) for histone marks and enhancers, F1 for promoters and for splice acceptors and donors, and accuracy for all splice sites. Learning rate and batch size per task are those Bio-xLSTM selected for DNA-xLSTM-PH (its Table A5). On splice acceptors and donors these values made GBST-BERT collapse to one class (F1 0.663 in all first three donor runs), so $1\mathrm{e}{-3}$ and $5\mathrm{e}{-4}$ were also run for both models with 3 seeds, and the setting with the best mean *validation* score was completed to 10 seeds.

Genomic Benchmarks. These 8 tasks [Grešová et al., 2023] add mouse and worm DNA, open chromatin and coding vs. non-coding sequence, and have up to 231k training sequences (Appendix B). The procedure is the same, with 10 epochs, 5 seeds and top-1 accuracy, as in the paper. It reports no per-task settings for this benchmark, so all tasks use one setting from its grid (learning rate $1\mathrm{e}{-3}$, batch 128). Sequences are cut to 1,024 bases, and shorter ones are left-padded; as in the repository's code, padded positions are not masked in attention or pooling.

Enhancer diagnosis. For Section 5.4, we compare validation and test scores over all tasks and train a logistic regression on the frequencies of the 256 possible 4-mers (no neural network, no pretraining) on 90% of a training file, scoring it on the remaining 10% and on the test file. We also count test sequences with an exact copy in the training file, and check their labels.

Reporting. As in Bio-xLSTM, we give the mean $\pm$ the range (maximum minus minimum) over seeds. Means within 0.0005 count as ties. The test sets were never used to choose a setting.

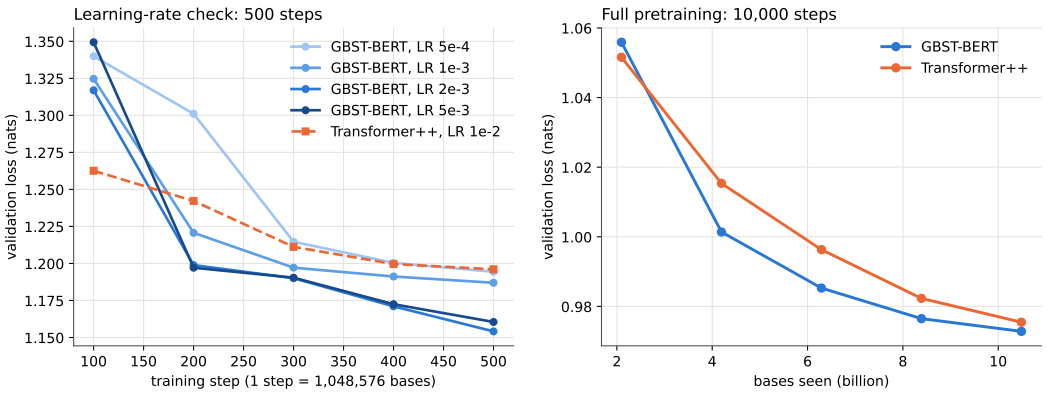


Figure 1: Validation loss. Left: learning-rate search for GBST-BERT over 500 steps, with Transformer++ at its paper value. Right: full pretraining of both models.

Table 3: Masked-base prediction on 2,048 unseen test sequences (311,767 masked bases, identical masks for both models).

Model	Test loss (nats)	Accuracy
GBST-BERT	**0.9753**	**56.44%**
Transformer++	0.9774	56.33%
Random guess	1.3863	25.00%
Always the most common base	–	29.42%

5 Results

5.1 Pretraining

Both models learn well beyond base composition (Table 3). GBST-BERT is ahead on validation loss for most of training (Figure 1), but the final gap of 0.11 percentage points cannot be interpreted with one run per model. Pretraining does not separate them.

5.2 What GBST learned

We measure the share of GBST's weight on each block size, averaged over 8 hg38 sequences from different chromosomes. For blocks of 1/2/3/4 bases it is 72/22/5/1% after 500 steps at learning rate $5e{-}4$, but 98/2/0/0% after 500 steps at $2e{-}3$ and in the final model. GBST therefore does *not* learn multi-base units on human DNA under this setup: at the learning rate we selected, it reads single bases almost exclusively. Each base has already been mixed with its neighbours by the convolution, so the module acts mainly as a learned local filter. The early difference between learning rates also shows that this outcome depends on training conditions.

5.3 Nucleotide Transformer benchmark

GBST-BERT has the best mean on 13 of 18 tasks (Table 4; xLSTM-PH 4, Mamba-PH 2, Mamba-PS 1, Transformer++ 0). It exceeds Transformer++ on 16 of 18 tasks (two-sided sign test over tasks $p = 0.0013$; the tasks are related, so this value is only a guide). The largest gains are on histone marks (e.g. H3K4me3 $+0.190$, H3K4me2 $+0.142$, H4ac $+0.114$ MCC) and on splice donors ($+0.071$ F1), while promoters differ by at most 0.005. Compared with the best paper model per task, GBST-BERT is higher on 12 tasks and tied on one. The splice tasks received an extra learning-rate check, which the paper's models did not need. Some GBST-BERT ranges are wide (H3K4me2 ± 0.212, H3K4me3 ± 0.180), so single runs are unreliable on these tasks.

Table 4: NT benchmark test scores, mean ± range over 10 seeds. The first five columns are Bio-xLSTM Table 1; the last two are ours. Bold: best mean (ties counted for each model).

Task	HyenaDNA	Mamba-PS	Mamba-PH	xLSTM-PS	xLSTM-PH	GBST-BERT	Transformer++
Histone marks (MCC)							
H3	0.779 ± 0.037	0.799 ± 0.029	0.815 ± 0.048	0.796 ± 0.014	$\mathbf{0.824 \pm 0.010}$	0.815 ± 0.031	0.801 ± 0.029
H3K14ac	0.612 ± 0.065	0.541 ± 0.212	0.631 ± 0.026	0.570 ± 0.008	0.598 ± 0.017	$\mathbf{0.673 \pm 0.016}$	0.569 ± 0.062
H3K36me3	0.613 ± 0.041	0.609 ± 0.109	0.601 ± 0.129	0.588 ± 0.019	0.625 ± 0.010	$\mathbf{0.655 \pm 0.035}$	0.591 ± 0.032
H3K4me1	0.512 ± 0.024	0.488 ± 0.102	0.523 ± 0.039	0.490 ± 0.012	0.526 ± 0.001	$\mathbf{0.565 \pm 0.038}$	0.481 ± 0.037
H3K4me2	0.455 ± 0.095	0.388 ± 0.101	0.487 ± 0.170	0.489 ± 0.024	0.504 ± 0.012	$\mathbf{0.509 \pm 0.212}$	0.367 ± 0.076
H3K4me3	0.549 ± 0.056	0.440 ± 0.202	0.544 ± 0.045	0.520 ± 0.019	0.537 ± 0.012	$\mathbf{0.553 \pm 0.180}$	0.362 ± 0.219
H3K79me3	0.672 ± 0.048	0.676 ± 0.026	$\mathbf{0.697 \pm 0.077}$	0.662 ± 0.011	$\mathbf{0.697 \pm 0.007}$	$\mathbf{0.697 \pm 0.132}$	0.676 ± 0.041
H3K9ac	0.581 ± 0.061	0.604 ± 0.048	0.622 ± 0.030	0.622 ± 0.013	0.627 ± 0.008	$\mathbf{0.648 \pm 0.037}$	0.599 ± 0.046
H4	0.763 ± 0.044	0.789 ± 0.020	0.811 ± 0.022	0.793 ± 0.011	$\mathbf{0.813 \pm 0.008}$	0.804 ± 0.074	0.803 ± 0.022
H4ac	0.564 ± 0.038	0.525 ± 0.240	0.621 ± 0.054	0.558 ± 0.018	0.583 ± 0.014	$\mathbf{0.642 \pm 0.037}$	0.528 ± 0.036
Enhancers (MCC)							
enhancers	0.517 ± 0.117	0.491 ± 0.066	$\mathbf{0.546 \pm 0.073}$	0.375 ± 0.030	0.545 ± 0.024	0.336 ± 0.085	0.368 ± 0.074
enhancers_types	0.386 ± 0.185	0.416 ± 0.095	0.439 ± 0.054	0.444 ± 0.046	$\mathbf{0.466 \pm 0.011}$	0.430 ± 0.167	0.384 ± 0.053
Promoters (F1)							
promoter_all	0.960 ± 0.005	0.967 ± 0.004	0.970 ± 0.004	0.962 ± 0.002	0.967 ± 0.001	$\mathbf{0.971 \pm 0.003}$	0.968 ± 0.006
promoter_no_tata	0.959 ± 0.011	0.968 ± 0.006	0.968 ± 0.010	0.963 ± 0.002	0.970 ± 0.001	$\mathbf{0.972 \pm 0.006}$	0.969 ± 0.004
promoter_tata	0.944 ± 0.040	$\mathbf{0.957 \pm 0.015}$	0.953 ± 0.016	0.948 ± 0.006	0.952 ± 0.005	0.948 ± 0.021	0.953 ± 0.013
Splice sites (accuracy / F1)							
splice_sites_all	0.956 ± 0.011	0.927 ± 0.021	0.940 ± 0.027	0.965 ± 0.006	0.974 ± 0.004	$\mathbf{0.980 \pm 0.004}$	0.975 ± 0.006
splice_sites_acceptors	0.958 ± 0.010	0.936 ± 0.077	0.937 ± 0.033	0.970 ± 0.005	0.953 ± 0.008	$\mathbf{0.977 \pm 0.009}$	0.955 ± 0.049
splice_sites_donors	0.949 ± 0.024	0.874 ± 0.289	0.948 ± 0.025	0.962 ± 0.004	0.951 ± 0.005	$\mathbf{0.980 \pm 0.012}$	0.909 ± 0.117
Tasks with the best mean	0	1	2	0	4	13	0

Table 5: Validation and test scores (MCC). Our models: mean over 10 seeds. 4-mer baseline: logistic regression on 4-mer frequencies, one 90/10 split. Last column: test sequences with an exact copy in the training file. Copies always carry the same label.

	GBST-BERT		Transformer++		4-mer logistic regression			Test seqs.
Task	val.	test	val.	test	val.	test	test, no copies	in train
enhancers	0.871	0.336	0.899	0.368	0.794	0.436	0.319	100/400
enhancers_types	0.789	0.430	0.788	0.384	0.777	0.299	0.168	100/400
promoter_all	0.973	0.971	0.970	0.968	0.807	0.802	0.802	21/5920
H3	0.811	0.815	0.787	0.801	0.634	0.631	0.631	0/1497

5.4 Why the enhancer tasks are weak

Enhancers are the clear weakness: MCC 0.336 against 0.546 for the best paper model. Our logs show that these two tasks stand apart: validation scores far exceed test scores (gap 0.53 and 0.36 for GBST-BERT), while on the 16 other tasks the gap never exceeds 0.05. Transformer++ behaves the same way, so GBST is not the cause.

To test whether this is a modelling problem at all, we trained a plain logistic regression on 4-mer frequencies, with no neural network and no pretraining (Table 5). It shows the same drop on the enhancer tasks (0.79 to 0.44, and 0.78 to 0.30) and none on the controls. The gap is therefore a property of the dataset: what separates the classes in the training file transfers only partly to the test file. On the binary task this simple baseline even exceeds both of our models on test, which suggests that they fit training-specific patterns more closely than a 4-mer model can.

Two further points affect how these tasks should be read. First, 100 of the 400 test sequences are exact copies of training sequences, so the test score partly measures recall; the baseline's score falls further when they are removed. Second, validation scores cannot guide model selection here, as they do not reflect test behaviour. We did not determine why some published models reach higher test scores on these tasks; with 400 test sequences the estimates are also noisy.

5.5 Genomic Benchmarks

Here the effect is much weaker (Table 6). GBST-BERT has the higher mean than Transformer++ on 5 of 8 tasks (sign test $p = 0.73$), by at most 0.8 percentage points. It is ahead on all 5 splits on coding

Table 6: Genomic Benchmarks test accuracy, mean ± range over 5 seeds. Paper models have 500k parameters and ours 2M, so the paper's values are a reference only. Last column: splits on which GBST-BERT is above Transformer++.

Task	Bio-xLSTM Table A2 (500k)				Ours (2M)		GBST > T++
	Mamba-PH	xLSTM-PH	Mamba-PS	xLSTM-PS	GBST-BERT	Transformer++	(splits)
Mouse enhancers	0.754 ± 0.074	0.780 ± 0.018	$\mathbf{0.793 \pm 0.058}$	0.778 ± 0.007	0.761 ± 0.116	0.776 ± 0.041	4/5
Coding vs. intergenomic	0.915 ± 0.003	0.931 ± 0.001	0.910 ± 0.003	0.934 ± 0.002	$\mathbf{0.937 \pm 0.002}$	0.930 ± 0.007	5/5
Human vs. worm	$\mathbf{0.973 \pm 0.001}$	0.965 ± 0.001	0.968 ± 0.002	0.956 ± 0.001	0.971 ± 0.003	0.972 ± 0.002	1/5
Human enhancers (Cohn)	$\mathbf{0.747 \pm 0.004}$	0.742 ± 0.005	0.745 ± 0.007	0.734 ± 0.005	0.738 ± 0.019	0.736 ± 0.014	3/5
Human enhancers (Ensembl)	0.893 ± 0.008	$\mathbf{0.920 \pm 0.001}$	0.900 ± 0.006	0.902 ± 0.004	0.903 ± 0.008	0.905 ± 0.008	1/5
Human regulatory	0.872 ± 0.011	0.872 ± 0.002	0.873 ± 0.007	0.869 ± 0.005	$\mathbf{0.885 \pm 0.003}$	0.883 ± 0.004	4/5
Human open chromatin	0.828 ± 0.006	0.826 ± 0.002	0.818 ± 0.006	0.800 ± 0.002	$\mathbf{0.841 \pm 0.009}$	0.836 ± 0.008	5/5
Human non-TATA promoters	0.946 ± 0.007	0.951 ± 0.004	0.945 ± 0.010	0.949 ± 0.001	$\mathbf{0.955 \pm 0.015}$	0.952 ± 0.004	2/5
Tasks with the best mean	2	1	1	0	4	0	

vs. intergenomic and open chromatin, and behind on 4 of 5 splits on two tasks. Mouse enhancers (968 training sequences, of which 97 go to validation) shows a lower mean although GBST-BERT wins 4 of 5 splits: in one split an early epoch scored highest on validation and then reached only 0.68 on test. Overall this benchmark does not show a clear benefit from GBST. It uses larger datasets, up to 231k training sequences, and one untuned setting, and we cannot separate these factors.

6 Discussion and limitations

What can be claimed. GBST-BERT is a competitive small DNA model: on the NT benchmark it matches or exceeds published 2M models on most tasks. It improves on its control on the NT benchmark, clearly on histone marks, and only marginally on Genomic Benchmarks.

What cannot be claimed yet. We have not shown that *tokenization* causes the gain. GBST reads single bases, so learned multi-base units cannot explain it; we suspect the convolution, since local context suits motif-driven tasks, but this is a hypothesis. In addition, the two models differ in normalization and in how their pretraining learning rate was chosen. The decisive experiment is GBST-BERT with GBST replaced by an embedding table, under otherwise identical settings. The code supports it (`TOKENIZER=embedding`) but it was not run. A further variant keeping only the convolution would isolate its contribution.

Further limitations. Each model was pretrained once, so variation from pretraining is unmeasured and the fine-tuning repeats capture only part of the uncertainty. Fine-tuning settings came from a different architecture (DNA-xLSTM-PH) and were not tuned for ours, except on splice tasks. GBST-BERT's first 3,000 pretraining steps used a data loader with a bug that corrupted 0.35% of sequences (Appendix C). We tested only one scale and context length, and block sizes up to 4. Our results concern classification benchmarks, not any clinical use.

7 Conclusion

A learned tokenizer did not behave as motivated on human DNA: it chose single bases. Even so, the resulting model is competitive at 2M parameters and beats its control on most NT tasks, with a much smaller effect on a second benchmark. The next steps are the ablation above and a repeat of pretraining with other seeds. Separately, the two NT enhancer tasks should be interpreted with care, given their train–test mismatch and duplicated sequences.

Code and reproducibility

- Repository (branch `charna-gbst`):
  https://github.com/OMAR622-ai/dna-xlstm-charna
- Full notebook, executed, with every step, table and figure:
  https://github.com/OMAR622-ai/dna-xlstm-charna/blob/charna-gbst/notebooks/GBST_BERT_full_pipeline.ipynb

- Fine-tuned NT models:
  https://github.com/OMAR622-ai/dna-xlstm-charna/releases/tag/nt-finetuned-models

The notebook re-computes the data statistics, the final test of both pretrained models, the GBST analysis, all tables and figures, the enhancer diagnosis, and one fine-tuning job per benchmark. It loads the saved results of the original pretraining and fine-tuning runs; setting its switches to `True` repeats those long runs with the same scripts.

References

Hugo Dalla-Torre, Liam Gonzalez, Javier Mendoza-Revilla, Nicolas Lopez Carranza, Adam Henryk Grzywaczewski, Francesco Oteri, Christian Dallago, Evan Trop, Bernardo P. de Almeida, Hassan Sirelkhatim, et al. The nucleotide transformer: Building and evaluating robust foundation models for human genomics. *bioRxiv*, pages 2023–01, 2023.

Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. BERT: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics*, pages 4171–4186, 2019.

Katarína Grešová, Vlastimil Martinek, David Čechák, Petr Šimeček, and Panagiotis Alexiou. Genomic benchmarks: a collection of datasets for genomic sequence classification. *BMC Genomic Data*, 24(1):25, 2023.

Adrián Morales-Pastor, Raquel Vázquez-Reza, Miłosz Wieczór, Clàudia Valverde, Manel Gil-Sorribes, Bertran Miquel-Oliver, Álvaro Ciudad, and Alexis Molina. Character-level tokenizations as powerful inductive biases for RNA foundational models. *arXiv preprint arXiv:2411.11808*, 2024.

Eric Nguyen, Michael Poli, Marjan Faizi, Armin Thomas, Michael Wornow, Callum Birch-Sykes, Stefano Massaroli, Aman Patel, Clayton Rabideau, Yoshua Bengio, et al. HyenaDNA: Long-range genomic sequence modeling at single nucleotide resolution. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 36, pages 43177–43201, 2023.

Yair Schiff, Chia Hsiang Kao, Aaron Gokaslan, Tri Dao, Albert Gu, and Volodymyr Kuleshov. Caduceus: Bi-directional equivariant long-range DNA sequence modeling. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, volume 235, pages 43632–43648, 2024.

Niklas Schmidinger, Lisa Schneckenreiter, Philipp Seidl, Johannes Schimunek, Pieter-Jan Hoedt, Johannes Brandstetter, Andreas Mayr, Sohvi Luukkonen, Sepp Hochreiter, and Günter Klambauer. Bio-xLSTM: Generative modeling, representation and in-context learning of biological and chemical sequences. *arXiv preprint arXiv:2411.04165*, 2024.

Noam Shazeer. GLU variants improve transformer. *arXiv preprint arXiv:2002.05202*, 2020.

Jianlin Su, Murtadha Ahmed, Yu Lu, Shengfeng Pan, Wen Bo, and Yunfeng Liu. RoFormer: Enhanced transformer with rotary position embedding. *Neurocomputing*, 568:127063, 2024.

Yi Tay, Vinh Q. Tran, Sebastian Ruder, Jai Gupta, Hyung Won Chung, Dara Bahri, Zhen Qin, Simon Baumgartner, Cong Yu, and Donald Metzler. Charformer: Fast character transformers via gradient-based subword tokenization. In *International Conference on Learning Representations (ICLR)*, 2022.

A Settings

Table 7 lists the main settings and their sources.

Table 7: Main settings and their sources.

Model	256 dimensions, 4 blocks, 8 heads, SwiGLU hidden size 320, no norm or linear biases [Bio-xLSTM Table A3]
GBST	blocks of 1–4 bases with offsets, sum pooling, calibration, no downsampling [ChaRNABERT Sec. 3.2]; maximum block size 4 is our choice
Masking	15% of positions; 80% `[MASK]`, 10% random, 10% unchanged [Bio-xLSTM App. C.5]
Batch	1,024 sequences of 1,024 bases per step [Bio-xLSTM App. C.5]
Optimizer	AdamW, $\beta = (0.9, 0.95)$, weight decay 0.1, 10% warm-up, cosine decay [Bio-xLSTM Table A4]
Learning rate	GBST-BERT $2\mathrm{e}{-3}$ (our search); Transformer++ $1\mathrm{e}{-2}$ [Bio-xLSTM Table A3]
Precision	32-bit; our GPUs (GTX 1080 Ti, RTX 2080 Ti, Quadro GV100) have no bf16

B Tasks

Table 8: The 18 NT tasks. Learning rate and batch size: Bio-xLSTM Table A5, DNA-xLSTM-PH column.

Task	Group	Classes	Length	Train	Test	Score	LR	Batch
H3	histone mark	2	500	13,468	1,497	MCC	8e-4	128
H3K14ac	histone mark	2	500	29,743	3,305	MCC	6e-4	128
H3K36me3	histone mark	2	500	31,392	3,488	MCC	6e-4	64
H3K4me1	histone mark	2	500	28,509	3,168	MCC	8e-4	128
H3K4me2	histone mark	2	500	27,614	3,069	MCC	6e-4	64
H3K4me3	histone mark	2	500	33,119	3,680	MCC	8e-4	128
H3K79me3	histone mark	2	500	25,953	2,884	MCC	1e-3	128
H3K9ac	histone mark	2	500	25,003	2,779	MCC	4e-4	64
H4	histone mark	2	500	13,140	1,461	MCC	8e-4	64
H4ac	histone mark	2	500	30,685	3,410	MCC	4e-4	64
enhancers	enhancer	2	200	14,968	400	MCC	2e-3	512
enhancers_types	enhancer	3	200	14,968	400	MCC	2e-3	512
promoter_all	promoter	2	300	53,276	5,920	F1	4e-4	64
promoter_no_tata	promoter	2	300	47,767	5,299	F1	1e-3	128
promoter_tata	promoter	2	300	5,509	621	F1	3e-3	128
splice_sites_all	splice site	3	400	27,000	3,000	accuracy	8e-4	64
splice_sites_acceptors	splice site	2	600	19,961	2,218	F1	2e-3	128
splice_sites_donors	splice site	2	600	19,775	2,198	F1	3e-3	128

Table 9: The 8 Genomic Benchmarks tasks.

Task	Classes	Train	Test
Mouse enhancers	2	968	242
Coding vs. intergenomic	2	75,000	25,000
Human vs. worm	2	75,000	25,000
Human enhancers (Cohn)	2	20,843	6,948
Human enhancers (Ensembl)	2	123,872	30,970
Human regulatory	3	231,348	57,713
Human open chromatin	2	139,804	34,952
Human non-TATA promoters	2	27,097	9,034

C Differences from the papers and known issues

- **ChaRNABERT Eq. 1** describes a depthwise convolution, but its Table 9 counts $d \cdot M + d^2$ weights, which implies an additional pointwise layer. We include it.
- **ChaRNABERT Eq. 3** read literally assigns a position blocks that do not contain it. We use blocks containing the position, as the text on offsets describes.
- **ChaRNABERT Eq. 9** differs from standard SwiGLU, which the text cites. We use the standard form.

- **Attention.** We use PyTorch's scaled dot-product attention instead of FlashAttention 2, which our GPUs do not support; the attention computed is the same.
- **NT data loader.** The repository's loader mixed all 18 tasks after a change on Hugging Face (461,850 examples instead of e.g. 5,509); it now loads each task's own file.
- **Genome reader.** Parallel reader processes shared one open file and sometimes read wrong bases: 0.35% of sequences with 2 workers per GPU, as used in the first part of training. Wrong reads stayed inside their data split. After the fix, GBST-BERT resumed from step 3,000 and Transformer++ was trained from scratch, so only GBST-BERT saw affected data.
- **Genomic Benchmarks schedule.** The repository's dataset configuration gives each task's total size as its training length, so the cosine schedule is longer than the real training run and the learning rate does not reach its minimum. This is the same for both of our models.
- **Validation at step 10,000.** After resuming, GBST-BERT's counter ran one step ahead and skipped this check; it was measured afterwards on the final model with the same settings.